Common theory for projective string markings

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Abstract

We give one reduction for projective string markings, without assuming the intersection property. Proper ends are controlled by the common centralizer of two involutions. A separate finite theorem shows that, when both proper end groups are S_4 or A_5 and both outer labels are odd, the marking gives precisely the 11-cell, the 57-cell, or the 4-simplex. The local enumeration, universal presentations and geometric conclusions are proved once and serve both polytope classifications and the later maniplex application. A half-turn extension theorem supplies the common construction for full and maximal ends. Dickson's subgroup classification, standard projective group structure and the group–polytope correspondence are the classical results used below.

1 Markings and classical results

A *string marking* of a group G is a generating ordered tuple $(r_0, \dots, r_{n-1})$ of distinct nonidentity involutions with $r_i r_j = r_j r_i$ whenever $|i - j| > 1$. Put $G_I = \langle r_i : i \in I \rangle$, with $G_\varnothing = 1$. Its labels are $|r_0 r_1|, \dots, |r_{n-2} r_{n-1}|$. The marking is a string C-marking if, in addition,

$$G_I \cap G_J = G_{I \cap J} \qquad (I, J \subseteq \{0, \dots, n-1\}). \tag{1}$$

For rank four write $H = \langle r_0, r_1, r_2 \rangle$ and $J = \langle r_1, r_2, r_3 \rangle$ for its two ends. Throughout, $L(q) = \mathrm{PSL}_2(q)$ and $P(q) = \mathrm{PGL}_2(q)$, for a prime power q.

We use the following classical results, in exactly the forms stated below. Dickson's classification says that a proper subgroup of $L(q)$ or $P(q)$, for odd q, is cyclic, dihedral, elementary abelian of characteristic order, a subgroup of an affine group $V \rtimes k^\times$, one of A_4, S_4, A_5, or a conjugate of a standard subfield group $L(k)$ or $P(k)$. Only groups which actually embed are intended. In the subfield alternatives $k \subseteq \mathbb{F}_q$ and the embedding is the natural matrix embedding, not an arbitrary abstract group isomorphism. The possible same-field proper subgroup $L(q) < P(q)$ must be included. We use

$$|L(q)| = \frac{q(q^2-1)}{(2, q-1)}, \qquad |P(q)| = q(q^2-1),$$

the simplicity of $L(q)$ for $q \geq 4$, and $C_{P(q)}(L(q)) = 1$. The small identifications are $L(2) = P(2) \cong S_3$, $L(3) \cong A_4$, $P(3) \cong S_4$, $L(5) \cong A_5$ and $P(5) \cong S_5$. These subgroup and structural facts are classical; subgroup multiplicity formulas from the sources are not inputs.

We also use the correspondence between ordered string C-markings and regular abstract polytopes, including the following clauses. Rank-i faces are cosets of $G_{\{0,\dots,n-1\}\setminus\{i\}}$; reversal gives duality; a marking-preserving epimorphism of string C-groups induces a covering. The local markings $[3,5]_5$ and $[5,3]_5$ below are the usual antipodal icosahedral and dodecahedral quotients. These names of the local maps, and the identification of the simplex, are standard geometric interpretations. They do not replace the required verification of the actual markings.

All groups $L(q), P(q)$ are centre-free and directly indecomposable. For the simple groups this is immediate. To see the assertion for $P(q)$ when $q \geq 4$, let commuting subgroups A, B generate $P(q)$; both are normal. The subgroup $[L(q), A]$ lies in $L(q) \cap A$ and is normal in the simple group $L(q)$. If it

is nontrivial, $L(q) \leq A$, so B centralizes $L(q)$ and is trivial. Otherwise A centralizes $L(q)$ and is trivial. The small cases follow from the normal-subgroup lists $1, A_3, S_3$; $1, V_4, A_4$; and $1, V_4, A_4, S_4$. In any string marking of these groups every adjacent product has order at least three. Indeed, a commuting adjacent pair splits the generators into two nontrivial commuting generated subgroups. Their intersection is central, hence trivial, and the resulting direct product contradicts indecomposability. This argument does not assume (1).

2 Common centralizers

In a string marking, the fourth generator commutes with the first two. Their common centralizer therefore controls both the possible rank and the choice of a fourth generator.

Lemma 1. *In characteristic two, the centralizer of a nonidentity involution in $P(q)$ is abelian. Let $k \subseteq K$ be finite fields of odd characteristic, and let $a, b \in P(k)$ be involutions with $|ab| > 2$. Their common centralizer in $P(K)$ lies in $P(k)$. It is trivial when ab is unipotent and has order two otherwise. In the latter case its nonidentity element is represented by the invertible matrix*

$$Z = A - \tfrac{1}{2}\operatorname{tr}(A)I, \tag{2}$$

where $A \in \mathrm{GL}_2(k)$ represents ab.

Proof. In characteristic two an involution is unipotent. Over an algebraic closure conjugate it to $x \mapsto x+1$. Its unique fixed point is infinity, so a commuting projectivity is $x \mapsto cx+d$. Commutation gives $c = 1$; these translations form an abelian group. The centralizer over the original field is a subgroup of this group.

In odd characteristic first suppose $u = ab$ is semisimple. Over the algebraic closure write $u : x \mapsto \lambda x$, with $\lambda \neq \pm 1$. A commuting projectivity preserves each of its two fixed points, since interchanging them would send u to $u^{-1} \neq u$. It is therefore $x \mapsto cx$. As $aua^{-1} = u^{-1}$, the involution a has the form $x \mapsto d/x$. Commutation with a gives $c^2 = 1$, so the common centralizer is $\{1, x \mapsto -x\}$. If A has eigenvalues α, β, the matrix in (2) has the nonzero eigenvalues $(\alpha - \beta)/2$ and $(\beta - \alpha)/2$. It represents $x \mapsto -x$ and has entries in k. Rescaling A only rescales Z, proving the claimed descent, including the nonsplit case.

If u is unipotent, conjugate it to $x \mapsto x+1$. Its centralizer again consists of translations $x \mapsto x+c$. An involution conjugating it to its inverse has the form $x \mapsto -x+d$. A commuting translation satisfies $2c = 0$, and hence $c = 0$. These cases exhaust u. □

Lemma 2. *Let G be centre-free with a string marking. Its first two generators do not commute. If every noncommuting pair of involutions has at most one nonidentity common centralizer element, the rank is at most four. If every involution centralizer is abelian, the rank is at most three.*

Proof. If r_0 commuted with r_1, it would commute with every generator by the string relations, so would be a nonidentity central element. At rank at least five, both r_3 and r_4 centralize r_0, r_1, contradicting the first centralizer hypothesis and their distinctness. Under the second hypothesis a rank-four marking puts r_0, r_1 in the abelian centralizer of r_3, again a contradiction. □

Lemma 3. *Let a rank-four string marking lie in $P(q)$, with q odd and adjacent labels greater than two. If H is proper, then $|r_0 r_1|$ is odd; if J is proper, then $|r_2 r_3|$ is odd.*

Proof. The nonidentity element r_3 centralizes $D = \langle r_0, r_1 \rangle$. Lemma 1 excludes the unipotent case and gives a unique nonidentity common centralizer element. If $m = |r_0 r_1|$ were even, $(r_0 r_1)^{m/2}$ would be that element. Then $r_3 \in H$, contrary to $G = \langle H, r_3 \rangle$ and properness. Reverse the indices for J. □

Theorem 4 (Projective reduction)**.** *A string marking of $L(q)$ or $P(q)$ has rank at most four; rank four requires odd q. In rank four every proper end of $L(q)$ is S_4 or A_5. Every proper end of $P(q)$ is S_4, A_5, or its natural subgroup $L(q)$. If both ends are proper, both are S_4 or A_5 and both outer labels are odd. These statements require no intersection property.*

Proof. The rank claims follow from Lemmas 1 and 2, since the projective groups are centre-free. This also handles $q = 2$; $L(3)$ has no generating tuple of involutions, since its involutions generate only V_4. All adjacent labels in the remaining cases exceed two by the first section.

Let $H = \langle r_0, r_1, r_2 \rangle$ be proper and put $D = \langle r_0, r_1 \rangle$. A cyclic or elementary abelian group cannot contain the nonabelian group D. Involutions in A_4 generate only V_4. In an affine group in odd characteristic every involution is $x \mapsto -x + v$; two such involutions commute only when they coincide. Thus the distinct commuting endpoints r_0, r_2 exclude this alternative, including affine dihedral groups of order $2p$.

If H is dihedral, its distinct commuting involutions r_0, r_2 force its rotation subgroup to have even order. (In an odd dihedral group two reflections commute only when equal.) Its central involution z and r_3 both centralize D. Lemma 1 gives $r_3 = z$, so $r_3 \in H$, a contradiction. This exclusion uses properness, not an intersection identity inside H.

If H is a natural subfield group, conjugate it into $P(k)$. Lemma 1 puts r_3 in the same $P(k)$, hence puts all of G there. For a proper subfield of size s we have $q \geq s^2$, and

$$|G| \geq q(q^2 - 1)/2 > s(s^2 - 1) = |P(k)|,$$

a contradiction. The only remaining same-field proper alternative is $H = L(q) < P(q)$. This proves the asserted lists of proper ends. The qualification "natural" is essential: exceptional $A_5 \cong L(5)$ and $S_4 \cong P(3)$ in other characteristics are not subfield exclusions.

Suppose now that both ends are proper. Lemma 3 makes both outer labels odd. If $H = L(q) < P(q)$, then H is normal of index two; $r_2 \in H$ and $r_3 \notin H$. The image of $r_2 r_3$ in $P(q)/H$ therefore has order two, impossible for its odd order. Reversal excludes the same alternative for J. The remaining ends are S_4, A_5. □

3 End groups isomorphic to S_4 or A_5

For a triple $f = (m, n, k)$ write

$$[m, n]_k = \langle x, y, z \mid x^2 = y^2 = z^2 = (xz)^2 = (xy)^m = (yz)^n = (xyz)^k = 1 \rangle. \tag{3}$$

The third parameter is the Petrie order, not an adjacent label. The six ordered local classes are

$$\begin{array}{c|ccc} S_4 & (3,3,4) & (3,4,3) & (4,3,3) \\ A_5 & (3,5,5) & (5,3,5) & (5,5,3). \end{array} \tag{4}$$

For $f = (m, n, k)$ and $v = (n, s, l)$ in this list define

$$\begin{aligned} U(f, v) = \langle r_0, r_1, r_2, r_3 \mid r_i^2 = 1,\ (r_i r_j)^2 = 1\ (j - i > 1), \\ (r_0 r_1)^m = (r_1 r_2)^n = (r_2 r_3)^s = 1, \\ (r_0 r_1 r_2)^k = (r_1 r_2 r_3)^l = 1 \rangle. \end{aligned} \tag{5}$$

Matching the middle label gives fourteen ordered pairs. Reversal sends (f, v) to (v^*, f^*), where $(a, b, c)^* = (b, a, c)$. There are ten reversal orbits, six fixed. Their complete table is:

Row	f	v	Order	Group	C-marking	Outer labels
1	$(5,5,3)$	$(5,5,3)$	1	trivial	no	$5,5$
2	$(5,5,3)$	$(5,3,5)$	1	trivial	no	$5,3$
3	$(5,3,5)$	$(3,5,5)$	3420	$L(19)$	yes	$5,5$
4	$(5,3,5)$	$(3,4,3)$	60	A_5	no	$5,4$
5	$(5,3,5)$	$(3,3,4)$	1	trivial	no	$5,3$
6	$(4,3,3)$	$(3,4,3)$	96	$2^4 : S_3$	yes	$4,4$
7	$(4,3,3)$	$(3,3,4)$	24	S_4	no	$4,3$
8	$(3,5,5)$	$(5,3,5)$	660	$L(11)$	yes	$3,3$
9	$(3,4,3)$	$(4,3,3)$	1	trivial	no	$3,3$
10	$(3,3,4)$	$(3,3,4)$	120	S_5	yes	$3,3$

Here an entry "yes" means that the specified generators, not merely some marking of the abstract group, form a string C-group. Rows which collapse do not have the indicated local groups; the local triples label their presentation input. The order-96 row has adjacent outer labels $4,4$, not $3,3$.

Lemma 5 (Exact finite certificate)**.** *Every rank-three string marking of S_4 or A_5 has the intersection property. Up to the full automorphism group its ordered type and Petrie order are exactly* (4)*;* (3) *is an exact presentation in each case. The ten-row table above is correct, including its groups, specified markings, collapse assertions and reversal coverage. In rows* $3,8,10$*, reversal extends to an automorphism of the marked group.*

Proof. The complete program `required_finite.g` enumerates all involutions in S_4, A_5, all ordered pairs of distinct commuting endpoints x, z, and every middle involution y. Its admission tests are only distinctness and generation. It admits 72 and 360 triples respectively. For every admitted triple it then checks both adjacent orders and all 64 subset-pair identities in (1). The intersection property is therefore proved for every admitted string marking; it is not assumed during the enumeration. The full automorphism orbits give the three classes in each row of (4). The code constructs each literal presentation (3) and computes its order 24 or 60. The marking provides a surjection onto the corresponding finite group; equality of orders makes it an isomorphism.

The code independently generates all fourteen compatible pairs, partitions by reversal and checks the ten supplied representatives and six fixed pairs. For each literal presentation (5), it computes the finite presentation order, constructs an isomorphic permutation group and transports the four original generators. It checks generator orders and distinctness, then tests IP over all sixteen subsets. Every accepted marking satisfies all 256 subset-pair identities; a rejected row may fail a generator condition or terminate at its first failed intersection. Thus the negative entries certify failure of the complete string C-marking conditions, not necessarily a separate IP test. Rows $3,8,10$ are identified by actual group isomorphisms with $L(19), L(11), S_5$; generator reversal is checked by constructing the homomorphism and testing bijectivity. In rows $4,7$ it verifies the indicated group and that one end is the whole group. In row 6 it verifies an elementary abelian normal subgroup of order 16, quotient S_3 and a complement. The four trivial rows are checked directly. These exact computations prove the finite assertions using GAP's algorithms for finite groups and finitely presented groups; they are not Lean kernel computations. □

Theorem 6 (Exceptional ends and odd outer labels)**.** *Let $(G; r_0, r_1, r_2, r_3)$ be a string marking whose two ends are proper and isomorphic to S_4 or A_5. If both outer labels are odd, then, up to isomorphism preserving the ordered marking, it is exactly one of rows* $3,8,10$*. Each row supplies one class, satisfies IP, and has type* $\{5,3,5\}$*,* $\{3,5,3\}$ *or* $\{3,3,3\}$ *respectively. In particular the statement does not require G to be a projective group.*

Proof. The local census and its exact presentations give a marking-preserving surjection from one of the ten groups $U(f, v)$ to G, after possibly reversing the marking. It is necessary to examine quotients, not just the universal groups themselves. The trivial rows cannot give nonidentity markers. In rows $4,7$, an end already equals the full universal group; this equality survives every quotient, contradicting

properness. Row 6 has first and last local product orders four, whereas the given ends have their exact local markings and hence these exact orders. It is excluded by oddness.

The remaining universals are the simple groups $L(19), L(11)$ and S_5. The simple groups have no nontrivial proper quotient. The normal subgroups of S_5 are $1, A_5, S_5$, so its only nontrivial proper quotient is C_2, which cannot have an end S_4 or A_5. Thus every surviving map is an isomorphism on all four distinguished generators. Lemma 5 gives IP and the asserted types. The local classes were taken under full automorphisms, so no marking was lost through a conjugacy restriction. The reversal automorphisms verified below identify the two representatives of each reversal orbit. Consequently each surviving row gives one class under isomorphisms preserving the ordered generators. □

Sharpness of the parity hypothesis. The verified order-96 row has proper S_4 ends, type $\{4, 3, 4\}$ and IP. It therefore contradicts the conclusion of three cases if the requirement that both outer labels be odd is removed. We make no classification of all quotients without that hypothesis.

Corollary 7. *Every string marking of $L(q)$ or $P(q)$ of rank at least four with both ends proper is one of these three markings. The first two occur in $L(19), L(11)$ respectively; the third occurs in $P(5)$. A string C-marking automatically has both ends proper, so the same conclusion applies to all projective string C-markings of rank at least four.*

Proof. Theorem 4 supplies rank four, the exceptional ends and odd outer labels; apply Theorem 6. The order formulas are strictly increasing in odd q. The groups $L(11), L(19)$ are simple, whereas odd-characteristic $P(q)$ has a proper subgroup of index two. The group S_5 is not simple and has order $120 = |P(5)|$; the small $L(3)$ is not S_5. These observations give exactly the stated field values and projective types. Finally, if an omitted generator belonged to its complementary subgroup, (1) would intersect that subgroup with its generator subgroup nontrivially. Hence C-markings have proper ends. □

4 Geometry and universal coverings

Theorem 8. *The three markings in Theorem 6 give the 57-cell, the 11-cell and the 4-simplex. Their types, local maps and face vectors are*

group	*type*	*facet / vertex figure*	(f_0, f_1, f_2, f_3)
$L(19)$	$\{5, 3, 5\}$	*hemi-dodecahedron / hemi-icosahedron*	$(57, 171, 171, 57)$
$L(11)$	$\{3, 5, 3\}$	*hemi-icosahedron / hemi-dodecahedron*	$(11, 55, 55, 11)$
S_5	$\{3, 3, 3\}$	*tetrahedron / tetrahedron*	$(5, 10, 10, 5)$.

All three are self-dual and universal among regular polytopes with their respective facet and vertex-figure types. The 11-cell edge graph is K_{11}. The even-word subgroup is the full group in the two simple rows, and is A_5 in the simplex row.

Proof. The two simple rows have local presentations $[3, 5]_5, [5, 3]_5$, with the standard antipodal map interpretations. Their exact global presentations are the string Coxeter relations of types $\{3, 5, 3\}$ and $\{5, 3, 5\}$, together with

$$(r_0 r_1 r_2)^5 = (r_1 r_2 r_3)^5 = 1.$$

For the third row send the four generators to $(12), (23), (34), (45)$. These generate S_5 and satisfy every relator in (5), including the two fourth powers of consecutive triple products. Both groups have order 120, so this is an isomorphism of marked groups with the standard simplex marking.

The program verifies that generator reversal extends to an automorphism in each of the three rows; the group–polytope correspondence gives self-duality. It also computes each of the four complementary-subgroup indices, giving the displayed face vectors. In the simple rows this has the following direct interpretation. Both ends have order 60. An edge stabilizer is $\langle r_0 \rangle \times \langle r_2, r_3 \rangle$, since the factors commute and IP makes their intersection trivial. Its order is $4m$ for outer label $m = 3$ or

5; the two-face stabilizer has the same order. Thus $660/60 = 11$, $660/12 = 55$, $3420/60 = 57$ and $3420/20 = 171$ give the first two vectors. For the 11-cell the code constructs actual vertex and edge cosets and checks their incidences: every edge has two endpoints and the 55 resulting unordered pairs are distinct. There are exactly 55 pairs on eleven vertices, proving K_{11} rather than inferring it merely from face counts.

The even-word subgroup is normal of index at most two: in its quotient all involutions have the same image. Simplicity makes it the full group in the two simple rows. In the simplex marking parity is the ordinary sign homomorphism, giving A_5.

Let Q be any regular polytope with one of the three indicated pairs of local maps. Its distinguished generators satisfy the corresponding literal presentation (5). There is therefore a marked epimorphism from the already verified string C-group of that row to its automorphism group. The classical covering theorem gives a covering of Q by the corresponding polytope above. No finiteness assumption on Q is needed: the epimorphism itself forces its group to be finite. This is universality of these exact markings, not an assumption that every abstract amalgam has IP. □

5 The common extension construction

We use the standard torus structure in $L(q)$ for odd q: its split and nonsplit cyclic tori have orders $(q-1)/2, (q+1)/2$, with dihedral normalizers of twice these orders. The split torus consists of diagonal projectivities of square eigenvalue ratio. The nonsplit torus comes from multiplication on $\mathbb{F}_{q^2}$ modulo scalars, with the determinant- square condition. Reflections in the normalizer interchange the two eigenlines. We use these classical torus facts, without assuming maximality of these normalizers.

Theorem 9 (Half-turn extension)**.** *Let q be odd, choose $\alpha \in \{1, -1\}$ with $q \equiv \alpha \pmod 4$, and put $N = (q-\alpha)/2$. Let (a, b, c) be a rank-three string marking of $H \le L(q)$, with $|bc| > 2$ and $|ab| = m > 1$ an odd divisor of N. There is a unique nonidentity projectivity z centralizing a, b; it lies in $L(q)$ and (a, b, c, z) is a string marking of $\langle H, z\rangle$. If $H = L(q)$, this group remains $L(q)$. If H is maximal proper in $L(q)$ and $C_H(\langle a, b\rangle) = 1$, it is again $L(q)$ and its first end is exactly H.*

Proof. The element ab is semisimple, since m divides $q - \alpha$ and hence is prime to the characteristic. It has two eigenlines over at most a quadratic extension. The two torus orders are coprime consecutive integers; since $m > 1$ divides N, its torus has order N. Both a, b invert ab, so preserve its unordered eigenline pair and normalize this torus. The even cyclic torus has a unique involution z. It commutes with the torus and its inverting normalizer coset, so centralizes a, b. Lemma 1 proves its asserted uniqueness even in $P(q)$.

The dihedral group $\langle a, b\rangle$ has odd rotation order and trivial centre. Thus z does not belong to it and differs from a, b. It also differs from c, because it commutes with b whereas $|bc| > 2$. The four generators are distinct nonidentity involutions and have every required distant commutation. They generate $\langle H, z\rangle$ by construction. If $H = L(q)$ there is nothing further to prove. Under the second hypothesis $z \notin H$, because its membership would give a nonidentity element of the stated trivial internal centralizer. Maximality then gives $\langle H, z\rangle = L(q)$. □

The extension theorem concerns string markings; it does not assert IP for the extended object. The maniplex application proves the correspondence with regular maniplexes and hence shows that the first end describes the original map. It proves its constructions for every prime in the stated ranges and identifies the generators with those in the source by explicit conjugacy. These conclusions do not rely on a sample of finite cases.

6 Attribution and verification

The six local classes, ten universal groups and the projective polytope classifications are prior results of Leemans–Schulte [1, 2]; the 11-cell and 57-cell are the prior constructions of Grünbaum and Coxeter.

The rank bound and classification of proper end groups, proved without assuming the intersection property, and the torus extension recover results of Leemans–Toledo [3]. Our common proof combines projective centralizer descent with a finite theorem for proper end groups S_4 and A_5 with odd outer labels. That theorem derives the exact marked groups, their intersection property and uniqueness from one calculation. It supplies the common argument for the polytope and maniplex applications. We make no claim to be the first to use these proof methods. The classical subgroup types, embeddings, torus structure and geometric correspondences remain attributed to the previous literature.

The finite calculations rely on the complete program `required_finite.g`. Besides the local and universal calculations proved above, it records the exact A_5 involution-centralizer, $L(23)$ symmetric-end and self-dual $L(13)$ witnesses used for the literal source corrections in the third application. Their exact conclusions are stated there. The optional `validation.g` contains finite samples of uniform constructions and controls against weakening hypotheses. Those samples are not premises of the all-prime proofs, which are written in full in the application. No mathematical branch of the required program is excluded from its cost.

Existing Lean components verify the normalizer parity argument, the abstract string centralizer rank obstruction, matrix normal-form identities and a generic extension interface. They do not formalize Dickson's theorem, projective descent in full, the GAP table, torus structure or the geometric correspondence. The source ledger records this boundary. A successful compilation of those components is not a Lean proof of this whole paper. The companion applications state exactly which previous results they recover and which additional statements they leave aside. This edition changes proof sharing, not the number of sources or questions credited.

References

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[3] D. Leemans and M. Toledo, *Maniplexes with automorphism group* $\mathrm{PSL}(2, q)$, bound eleven-page author manuscript dated 15 May 2023.

[4] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge University Press, 2002; string C-group and covering correspondence.